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Tsuchiya

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(54) **CHARGING APPARATUS**

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B60L 53/35 (2019.01)
H02J 7/14 (2006.01)
B60L 53/16 (2019.01)

(52) **U.S. Cl.**

CPC ***B60L 53/35*** (2019.02); ***H02J 7/0013*** (2013.01); ***H02J 7/0042*** (2013.01); *B60L 53/16* (2019.02)

(58) **Field of Classification Search**

CPC B60L 53/35; B60L 53/38; B60L 53/39; B60L 53/34; B60L 53/31; B60L 53/30; H02J 7/0013; H02J 7/0042
USPC 320/132, 134
See application file for complete search history.

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(57) **ABSTRACT**

A charging apparatus is installed in a partition portion that separates a first parking space and a second parking space, and configured to charge an energy storage device mounted on a vehicle parked in one of the first parking space and the second parking space. The charging apparatus includes a movable portion that includes a connection device that is connectable to the energy storage device, and a moving device. The moving device is configured to move the movable portion to one of a first position at which the movable portion projects into a space that is close to the first parking space with respect to the partition portion, a second position at which the movable portion projects into a space that is close to the second parking space with respect to the partition portion, and a third position at which the movable portion is housed in the partition portion.

7 Claims, 7 Drawing Sheets

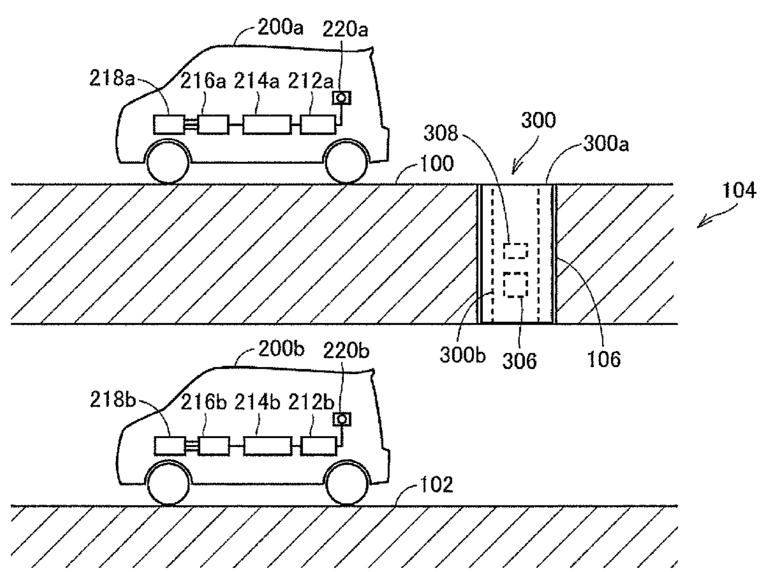

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FIG. 1

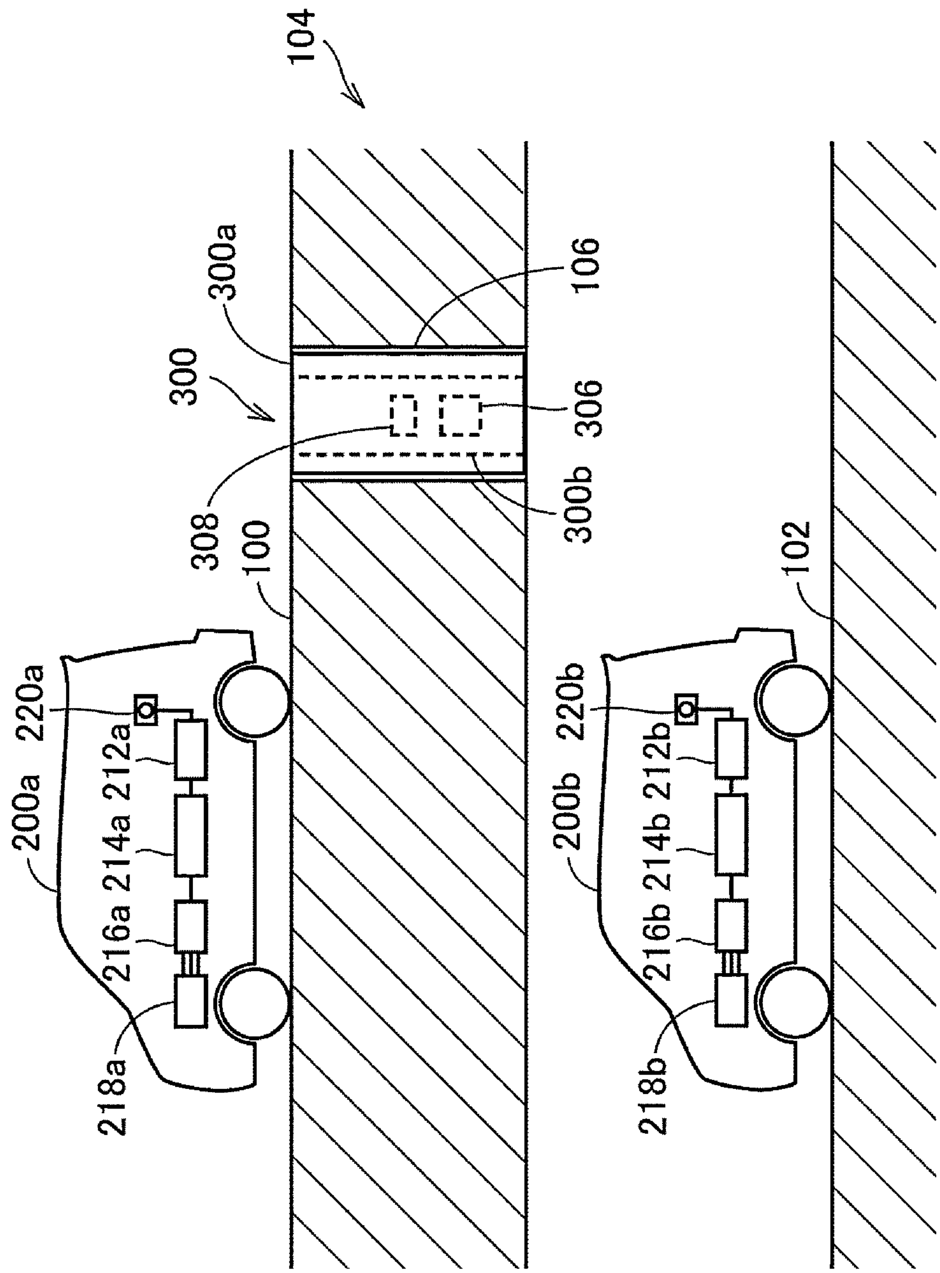

104
300a
106
300
306
308
300b
100
102
200a
220a
212a
214a
216a
218a
200b
220b
212b
214b
216b
218b

FIG. 2

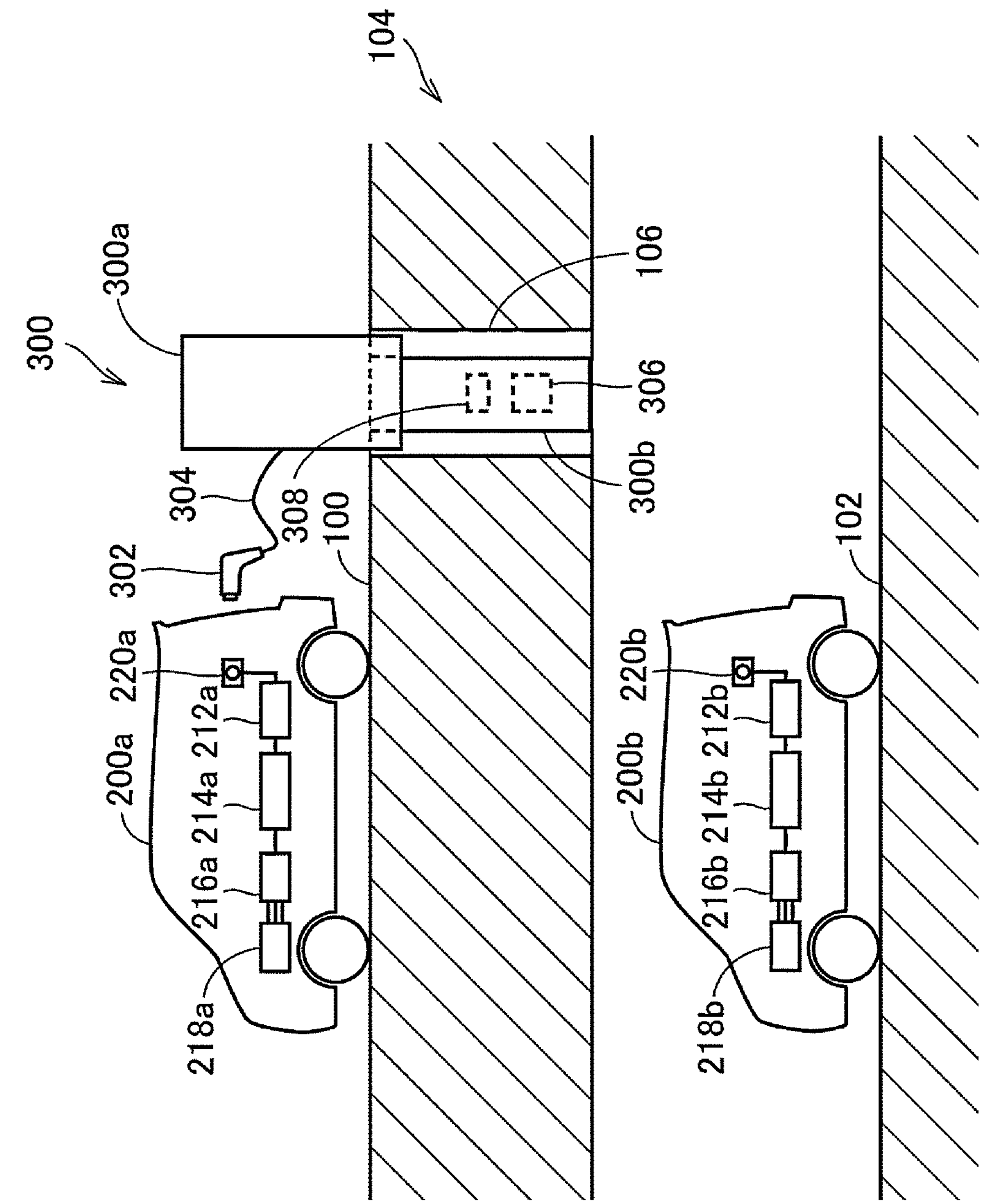

104
300
300a
106
306
308
300b
304
100
302
102
220a
212a
214a
216a
218a
200a
220b
212b
214b
216b
218b
200b

FIG. 3

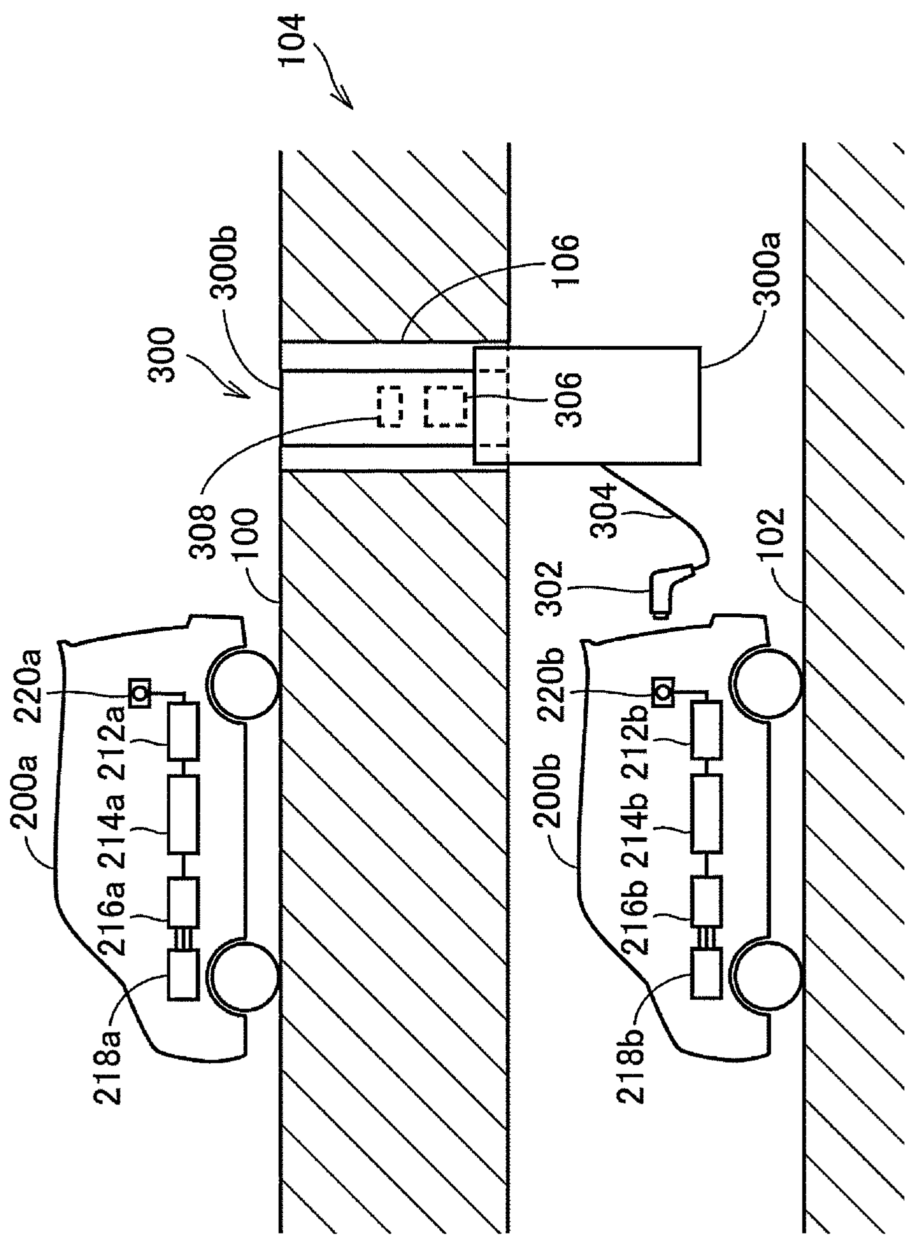

104
300b
106
300a
300
306
308
100
304
302
102
220a
200a
214a
212a
216a
218a
220b
200b
214b
212b
216b
218b

FIG. 4

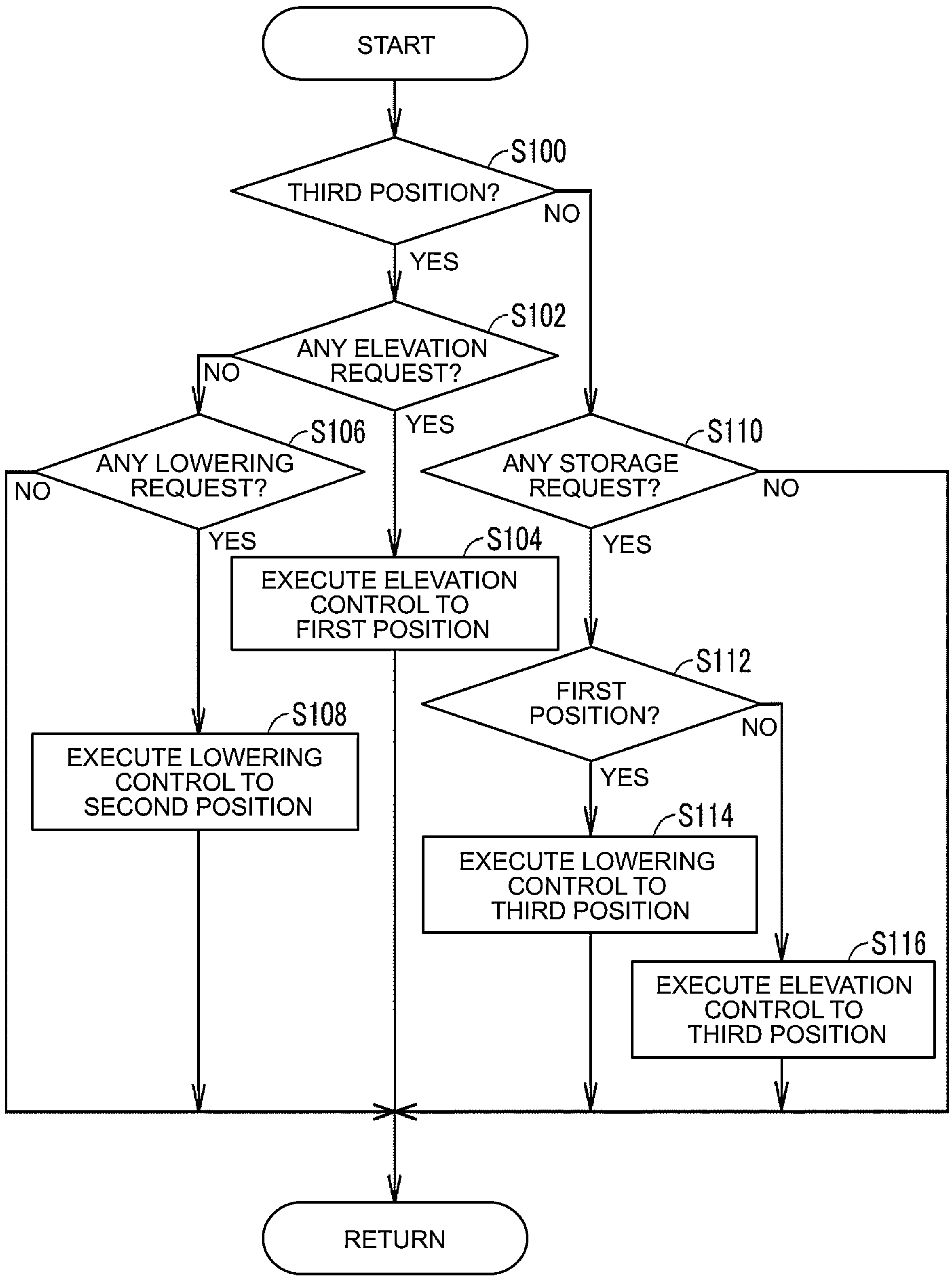

START
S100
THIRD POSITION?
NO
YES
S102
ANY ELEVATION REQUEST?
NO
YES
S106
ANY LOWERING REQUEST?
NO
YES
S110
ANY STORAGE REQUEST?
NO
YES
S104
EXECUTE ELEVATION CONTROL TO FIRST POSITION
S112
FIRST POSITION?
NO
YES
S108
EXECUTE LOWERING CONTROL TO SECOND POSITION
S114
EXECUTE LOWERING CONTROL TO THIRD POSITION
S116
EXECUTE ELEVATION CONTROL TO THIRD POSITION
RETURN

FIG. 5

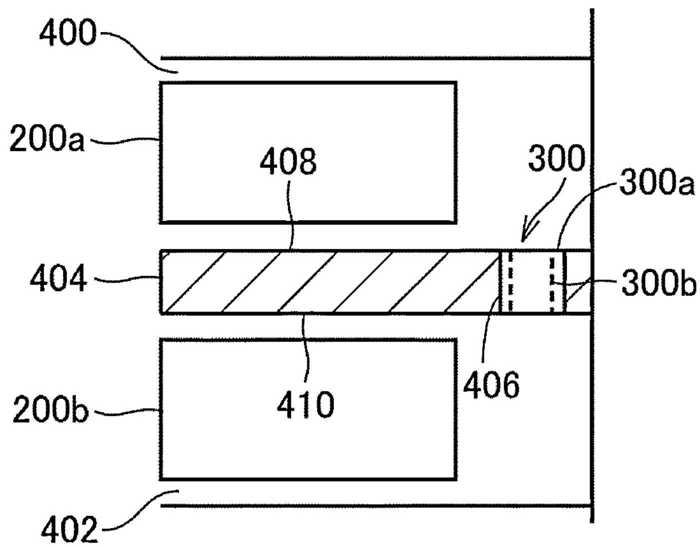

400
200a
408
300
300a
404
300b
406
410
200b
402

FIG. 6

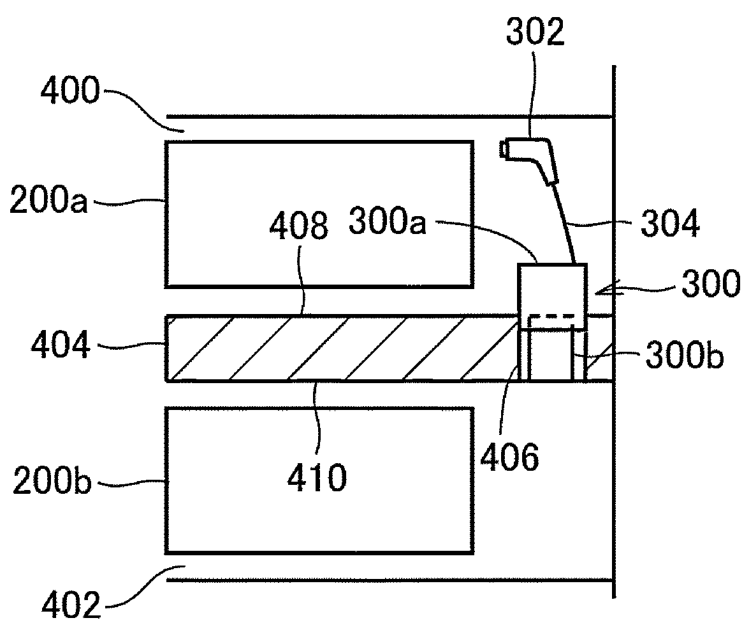

302
400
200a
408
300a
304
300
404
300b
406
410
200b
402

FIG. 7

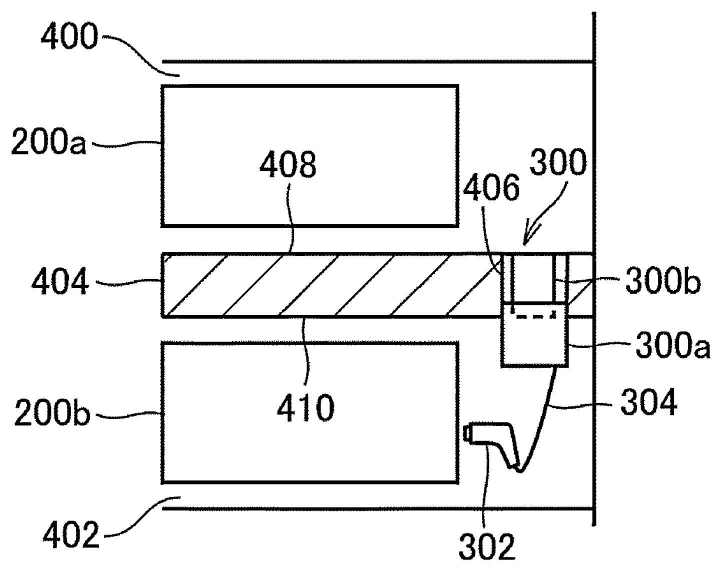
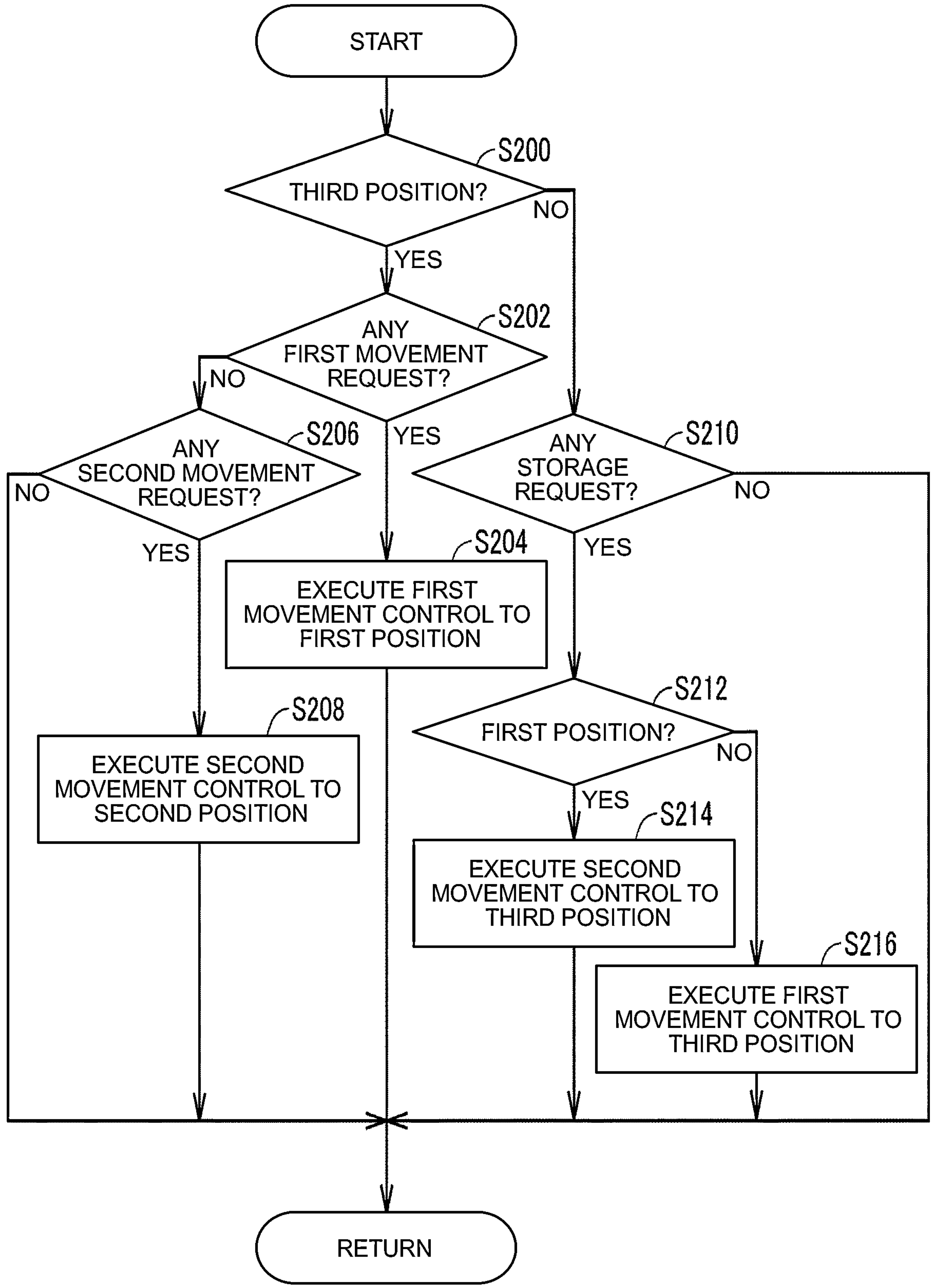


FIG. 8


START
S200
THIRD POSITION?
NO
YES
S202
ANY FIRST MOVEMENT REQUEST?
NO
YES
S206
ANY SECOND MOVEMENT REQUEST?
NO
YES
S204
EXECUTE FIRST MOVEMENT CONTROL TO FIRST POSITION
S208
EXECUTE SECOND MOVEMENT CONTROL TO SECOND POSITION
S210
ANY STORAGE REQUEST?
NO
YES
S212
FIRST POSITION?
NO
YES
S214
EXECUTE SECOND MOVEMENT CONTROL TO THIRD POSITION
S216
EXECUTE FIRST MOVEMENT CONTROL TO THIRD POSITION
RETURN

CHARGING APPARATUS

CROSS-REFERENCE TO RELATED APPLICATION

This application claims priority to Japanese Patent Application No. 2021-047078 filed on Mar. 22, 2021, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

The present disclosure relates to a charging apparatus configured to charge an energy storage device mounted on a vehicle.

2. Description of Related Art

A charging apparatus for charging an energy storage device mounted on a vehicle etc. is installed outside the vehicle etc., such as in a parking lot or on a sidewalk. However, such a charging apparatus may disturb traveling of vehicles or walking as it occupies an installation space. A technique of rendering the charging apparatus movable and housing the charging apparatus under the ground, for example, is known in the art.

Japanese Unexamined Patent Application Publication No. 2011-109807 (JP 2011-109807 A), for example, discloses a charging pole installed so as to be movable up and down, to be elevated to stand on the ground and lowered to be housed under the ground.

SUMMARY

A parking lot in which vehicles are parked occasionally includes parking spaces provided in a plurality of levels, such as when the parking lot is a multilevel parking lot or when parking spaces are provided both on and under the ground. Therefore, in order to enable electric vehicles to be charged in each level, it is conceivable to install the movable charging apparatus discussed above in each level, for example. However, it may be expensive to install the movable charging apparatus in each level. Further, it may also be expensive to install the movable charging apparatus in each of parking spaces in a case a large number of parking spaces are provided in a single-level parking lot.

The present disclosure provides a charging apparatus that suppresses an increase in the installation cost for installing the charging apparatus so as to be usable in a plurality of parking spaces.

A charging apparatus according to an aspect of the present disclosure is installed in a partition portion that separates a first parking space and a second parking space, and configured to charge an energy storage device mounted on a vehicle parked in one of the first parking space and the second parking space. The charging apparatus includes a movable portion that includes a connection device that is connectable to the energy storage device, and a moving device. The moving device is configured to move the movable portion to one of a first position at which the movable portion projects into a space that is close to the first parking space with respect to the partition portion, a second position at which the movable portion projects into a space that is close to the second parking space with respect to the partition portion, and a third position at which the movable portion is housed in the partition portion.

With such a configuration, the energy storage device mounted on the vehicle parked in either the first parking space or the second parking space can be charged using a single charging apparatus. Consequently, it is not necessary to install the charging apparatus in each of the first parking space and the second parking space. Therefore, it is possible to suppress an increase in the installation cost for installing the charging apparatus so as to be usable in a plurality of parking spaces.

In the charging apparatus according to the aspect of the present disclosure, the first parking space may be located in a level directly above a level of the second parking space. The partition portion may be provided between a floor of the first parking space and a ceiling of the second parking space.

With such a configuration, the energy storage device mounted on the vehicle parked in either the first parking space or the second parking space, which are located separately in the upper and lower levels, can be charged using a single charging apparatus. Consequently, it is not necessary to install the charging apparatus in each of the first parking space and the second parking space. Therefore, it is possible to suppress an increase in the installation cost for installing the charging apparatus so as to be usable in a plurality of parking spaces.

In the charging apparatus according to the aspect of the present disclosure, the first parking space may be adjacent to the second parking space in a horizontal direction. The partition portion may be provided between a partition wall of the first parking space and a partition wall of the second parking space.

With such a configuration, the energy storage device mounted on the vehicle parked in either the first parking space or the second parking space, which are adjacent in the horizontal direction, can be charged using a single charging apparatus. Consequently, it is not necessary to install the charging apparatus in each of the first parking space and the second parking space. Therefore, it is possible to suppress an increase in the installation cost for installing the charging apparatus so as to be usable in a plurality of parking spaces.

In the charging apparatus according to the aspect of the present disclosure, the charging apparatus may further include a control device configured to control the moving device using information acquired from an external device that is external to the charging apparatus. The control device may be configured to control the moving device so as to move the movable portion to one of the first position, the second position, and the third position corresponding to a movement request to move the movable portion when the movement request is acquired from the external device.

With such a configuration, the movable portion can be moved to one of the first position, the second position, and the third position in accordance with the movement request from the external device.

With the present disclosure, it is possible to provide a charging apparatus that suppresses an increase in the installation cost for installing the charging apparatus so as to be usable in a plurality of parking spaces.

BRIEF DESCRIPTION OF THE DRAWINGS

Features, advantages, and technical and industrial significance of exemplary embodiments of the present disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

FIG. **1** illustrates an example of the configuration of a charging station in a state in which a movable portion is housed in a partition portion between a first parking space and a second parking space and electric vehicles;

FIG. **2** illustrates an example of the configuration of the charging station with the movable portion located at a first position and the electric vehicles;

FIG. **3** illustrates an example of the configuration of the charging station with the movable portion located at a second position and the electric vehicles;

FIG. **4** is a flowchart illustrating an example of processes executed by a control device;

FIG. **5** illustrates an example of the configuration of a charging station in a state in which a movable portion is housed in a partition portion between a first parking space and a second parking space and electric vehicles according to a modification;

FIG. **6** illustrates an example of the configuration of the charging station with the movable portion located at a first position and the electric vehicles according to the modification;

FIG. **7** illustrates an example of the configuration of the charging station with the movable portion located at a second position and the electric vehicles according to the modification; and

FIG. **8** is a flowchart illustrating an example of processes executed by a control device according to the modification.

DETAILED DESCRIPTION OF EMBODIMENTS

An embodiment of the present disclosure will be described in detail below with reference to the drawings. The same or corresponding parts are denoted by the same signs throughout the drawings, and description thereof will not be repeated.

The configuration of a charging station **300** that is a charging apparatus according to an embodiment of the present disclosure will be described below by way of example. FIG. **1** illustrates an example of the configuration of the charging station **300** in a state in which a movable portion **300***a* is housed in a partition portion **104** between a first parking space and a second parking space and electric vehicles **200***a*, **200***b*.

In the state illustrated in FIG. **1**, the charging station **300** is configured such that the upper end of the charging station **300** is generally positioned at the same position as the floor surface of a first parking space **100** and the lower end of the charging station **300** is generally positioned at the same position as the ceiling surface of a second parking space **102**.

The charging station **300** has a housing in a cylindrical shape, and is installed in an opening portion **106** formed in a portion of the partition portion **104** between the floor surface of the first parking space **100** and the ceiling surface of the second parking space **102**.

The opening portion **106** is formed to penetrate the partition portion **104** in the vertical direction. The opening portion **106** has a circular cross-sectional shape, and is formed to have a predetermined gap between an inner peripheral surface of the opening portion **106** and an outer peripheral surface of the housing of the charging station **300**. Further, the opening portion **106** is formed so as to have a length that is about the same as the length of the charging station **300** in the vertical direction in the state illustrated in FIG. **1**.

The first parking space **100** and the second parking space **102** are located separately in upper and lower levels. More specifically, the first parking space **100** is located in the level directly above the level of the second parking space **102**.

The charging station **300** includes a movable portion **300***a* and a fixed portion **300***b*. A storage space (not illustrated) in which a connector **302** (see FIGS. **2** and **3**) can be housed is formed in the movable portion **300***a*. One end of a cable **304** (see FIGS. **2** and **3**) is connected to the connector **302**, while the other end of the cable **304** is connected to a power source (not illustrated). The power source may be an alternating-current (AC) power source such as a commercial power source, for example. The cable **304** has a shape-based expandable portion that has a curled portion or a structure-based expandable portion that has a winding structure, and is configured to be expandable to an inlet of an electric vehicle parked in either the first parking space **100** or the second parking space **102** when the connector **302** is taken out.

The movable portion **300***a* of the charging station **300** is configured to be movable to a first position at which the movable portion **300***a* projects into a space on the first parking space **100** side.

FIG. **2** illustrates an example of the configuration of the charging station **300** with the movable portion **300***a* located at the first position and the electric vehicles **200***a*, **200***b*. When the movable portion **300***a* of the charging station **300** is elevated to the first position on the first parking space **100** side, as illustrated in FIG. **2**, the connector **302** housed in the movable portion **300***a* can be taken out of the movable portion **300***a* to be connected to an inlet **220***a* of the electric vehicle **200***a* parked in the first parking space **100**.

Further, the movable portion **300***a* is configured to be movable to a second position at which the movable portion **300***a* projects into a space on the second parking space **102** side.

FIG. **3** illustrates an example of the configuration of the charging station **300** with the movable portion **300***a* located at the second position and the electric vehicles **200***a*, **200***b*. When the movable portion **300***a* of the charging station **300** is lowered to the second position on the second parking space **102** side, as illustrated in FIG. **3**, the connector **302** housed in the movable portion **300***a* can be taken out of the movable portion **300***a* to be connected to an inlet **220***b* of the electric vehicle **200***b* parked in the second parking space **102**.

The connector **302** is housed at a position at which the connector **302** can be taken out of the movable portion **300***a* to the inlet **220***a* of the electric vehicle **200***a* by a user situated in the level of the first parking space **100** when the movable portion **300***a* is located at the first position, for example. Further, the connector **302** is housed at a position at which the connector **302** can be taken out of the movable portion **300***a* to the inlet **220***b* of the electric vehicle **200***b* by a user situated in the level of the second parking space **102** when the movable portion **300***a* is located at the second position, for example.

While the connector **302** is described as being housed at a middle portion, in the up-down direction, of the movable portion **300***a* in the present embodiment, for example, the connector **302** is not specifically limited to being housed at a middle portion in the up-down direction.

For example, a first connector that can be taken out of the movable portion **300***a* by a user situated in the level of the first parking space **100** when the movable portion **300***a* is located at the first position may be housed at the upper end of the movable portion **300***a*. Further, a second connector that can be taken out of the movable portion **300***a* by a user situated in the level of the second parking space **102** when the movable portion **300***a* is located at the second position may be housed at the lower end of the movable portion **300***a*.

Further, the movable portion **300***a* is configured to be held at a third position at which the movable portion **300***a* is housed in the partition portion **104** as illustrated in FIG. **1**.

The fixed portion **300***b* is fixed to the inner wall portion of the opening portion **106**. The fixed portion **300***b* includes an elevating device **306** that elevates and lowers the movable portion **300***a* in the up-down direction and a control device **308** that controls operation of the elevating device **306**.

The elevating device **306** may have a rack-and-pinion mechanism that elevates and lowers the movable portion **300***a* by rotating a pinion gear meshed with a rack gear fixed to the movable portion **300***a* using an electric actuator, may have a mechanism that uses a hydraulic cylinder and that elevates and lowers the movable portion **300***a* by increasing and decreasing a hydraulic pressure to be supplied to a cylinder body fixed to the fixed portion **300***b* with a rod connected a piston fixed to the movable portion **300***a*, or may have a mechanism that elevates and lowers the movable portion **300***a* by generating a magnetic repulsive force between the movable portion **300***a* and the fixed portion **300***b*, for example.

The elevating device **306** is configured such that the movable portion **300***a* is not elevated beyond the first position or not lowered beyond the second position using a stopper mechanism etc., for example. The elevating device **306** may have a lock mechanism configured to limit the position of the movable portion **300***a* at the third position and cancel the limitation on the position of the movable portion **300***a*, for example. The elevating device **306** is an example of a moving device according to the present disclosure.

The control device **308** includes a central processing unit (CPU) and a memory composed of a read only memory (ROM), a random access memory (RAM), etc. The control device **308** controls an electric device (e.g. the elevating device **306**) provided in the charging station **300** based on information stored in the memory, information acquired from the outside of the charging station **300** using a communication device (not illustrated), and information acquired from various sensors. This control is not limited to software processing executed by the CPU, and may be constructed by dedicated hardware (an electronic circuit).

The charging station **300** is configured to be able to communicate various kinds of information with devices that are external to the charging station **300** using the communication device. The charging station **300** may be configured to be communicable with a management server (not illustrated) that manages a plurality of charging stations including the charging station **300**, may be configured to be communicable with a portable terminal (not illustrated) owned by a user, may be configured to be communicable with electric vehicles (the electric vehicles **200***a*, **200***b* in FIG. **1**) to be charged by the charging station **300**, or may be configured to be communicable with another charging station **300**, for example.

The control device **308** controls the elevating device **306** such that the movable portion **300***a* is elevated from the third position to the first position when a condition for executing elevation control is met, for example. The condition for executing the elevation control includes a condition that the movable portion **300***a* is located at the third position and a condition that there is a request to execute the elevation control (hereinafter referred to as an "elevation request"), for example. The elevation request may be received from the electric vehicle **200***a* parked in the first parking space **100**, or may be received from the management server, the portable terminal, or another charging station **300**, for example. The condition for executing the elevation control may include a condition that the movable portion **300***a* is located at the second position or the third position, in place of the condition that the movable portion **300***a* is located at the third position. The control device **308** detects the amount of elevation of the movable portion **300***a* using a sensor (not illustrated), and elevates the movable portion **300***a* until the amount of elevation of the movable portion **300***a* reaches a value that is equivalent to the first position, for example.

The control device **308** sets a first flag, which indicates that the movable portion **300***a* is located at the first position, to an on state when the movable portion **300***a* is located at the first position, for example. The control device **308** sets the first flag to an off state when the movable portion **300***a* is located at a position other than the first position, for example.

Alternatively, the control device **308** controls the elevating device **306** such that the movable portion **300***a* is lowered from the third position to the second position when a condition for executing lowering control is met, for example. The condition for executing the lowering control includes a condition that the movable portion **300***a* is located at the third position and a condition that there is a request to execute the lowering control (hereinafter referred to as a "lowering request"), for example. The lowering request may be received from the electric vehicle **200***b* parked in the second parking space **102**, or may be received from the management server, the portable terminal, or another charging station **300**, for example. The condition for executing the lowering control may include a condition that the movable portion **300***a* is located at the first position or the third position, in place of the condition that the movable portion **300***a* is located at the third position. The control device **308** lowers the movable portion **300***a* until the amount of elevation of the movable portion **300***a* reaches a value that is equivalent to the second position, for example.

The control device **308** sets a second flag, which indicates that the movable portion **300***a* is located at the second position, to an on state when the movable portion **300***a* is located at the second position, for example. The control device **308** sets the second flag to an off state when the movable portion **300***a* is located at a position other than the second position, for example.

Further, the control device **308** controls the elevating device **306** such that the movable portion **300***a* is moved from the first position or the second position to the third position when a condition for executing storage control is met. Specifically, the control device **308** executes the lowering control such that the movable portion **300***a* is lowered from the first position to the third position to be housed in the partition portion **104** when the movable portion **300***a* is located at the first position and the condition for executing the storage control is met, for example. Further, the control device **308** executes the elevation control such that the movable portion **300***a* is elevated from the second position to the third position to be housed in the partition portion **104** when the movable portion **300***a* is located at the second position and the condition for executing the storage control is met, for example. In addition, the control device **308** elevates and lowers the movable portion **300***a* until the amount of elevation of the movable portion **300***a* reaches a value that is equivalent to the third position, for example.

The control device **308** sets a third flag, which indicates that the movable portion **300***a* is located at the third position, to an on state when the movable portion **300***a* is located at the third position, for example. The control device **308** sets the third flag to an off state when the movable portion **300***a* is located at a position other than the third position, for example.

The condition for executing the storage control includes a condition that the movable portion **300***a* is located at the first position or the second position and a condition that there is a storage request to house the charging station **300**. The storage request may be received from the electric vehicle **200***a* parked in the first parking space **100** or the electric vehicle **200***b* parked in the second parking space **102** when the movable portion **300***a* is located at the first position, for example. Alternatively, the storage request may be received from the electric vehicle **200***a* parked in the first parking space **100** or the electric vehicle **200***b* parked in the second parking space **102** when the movable portion **300***a* is located at the second position, for example. Alternatively, the storage request may be received from the management server, the portable terminal, or another charging station **300**, for example.

FIGS. **1**, **2**, and **3** further illustrate an example of the configuration of the electric vehicle **200***a* parked in the first parking space **100** and the configuration of the electric vehicle **200***b* parked in the second parking space **102**, the electric vehicles **200***a*, **200***b* being chargeable by the charging station **300**.

As illustrated in FIGS. **1**, **2**, and **3**, the electric vehicles **200***a*, **200***b* include vehicles on which an energy storage device is mounted, such as plug-in hybrid vehicles and electric vehicles, for example. It is only necessary that the electric vehicles **200***a*, **200***b* should have components that enable reception of electric power supplied from the charging station **300**. In particular, the electric vehicles **200***a*, **200***b* are not limited to the vehicles mentioned above, and may be vehicles on which an energy storage device for external power supply is mounted, for example.

An example of the configuration of the electric vehicle **200***a* will be described below. The electric vehicle **200***a* includes a charger **212***a*, a battery **214***a*, an inverter **216***a*, a motor generator **218***a*, and an inlet **220***a*.

When AC electric power is supplied from the inlet **220***a*, the charger **212***a* converts the supplied AC electric power into direct-current (DC) electric power, and supplies the DC electric power to the battery **214***a*. The battery **214***a* is charged through operation of the charger **212***a*. The charger **212***a* is controlled in accordance with a control signal from an electronic control unit (ECU) (not illustrated) of the electric vehicle **200***a*, for example.

The battery **214***a* is a rechargeable electric power storage element, for example. A secondary battery such as a nickel metal hydride battery or a lithium ion battery containing a liquid or solid electrolyte is typically used as the battery **214***a*. Alternatively, the battery **214***a* may be an energy storage device that can store electric power. A large-capacity capacitance may be used in place of the battery **214***a*, for example.

The inverter **216***a* converts DC electric power from the battery **214***a* into AC electric power, and supplies the AC electric power to the motor generator **218***a*, for example. The inverter **216***a* also converts AC electric power (regenerated electric power) from the motor generator **218***a* into DC electric power, and supplies the DC electric power to the battery **214***a* to charge the battery **214***a*, for example.

The motor generator **218***a* receives electric power supplied from the inverter **216**, and supplies a rotational force to drive wheels. The drive wheels are rotated by the rotational force supplied by the motor generator **218***a*, and cause the electric vehicle **200***a* to travel.

The inlet **220***a* is provided in an exterior portion of the electric vehicle **200***a* together with a cover (not illustrated) such as a lid. The inlet **220***a* is a power reception portion that receives charging electric power supplied from an external charging apparatus (e.g. the charging station **300**). The inlet **220***a* is shaped to enable the connector **302** of the charging station **300** to be attached thereto. Both the inlet **220***a* and the connector **302** have built-in contacts. When the connector **302** is attached to the inlet **220***a*, the contacts contact each other to electrically connect the inlet **220***a* and the connector **302** to each other. At this time, the battery **214***a* of the electric vehicle **200***a* can be charged using electric power supplied from the charging station **300**.

The charger **212***b*, the battery **214***b*, the inverter **216***b*, the motor generator **218***b*, and the inlet **220***b* included in the electric vehicle **200***b* have the same configuration as the charger **212***a*, the battery **214***a*, the inverter **216***a*, the motor generator **218***a*, and the inlet **220***a*, respectively, included in the electric vehicle **200***a*. Therefore, detailed description thereof will not be repeated.

A charging apparatus for charging the batteries **214***a*, **214***b* mounted on the electric vehicles **200***a*, **200***b* discussed above is installed in a parking lot or on a sidewalk, for example. However, such a charging apparatus may disturb traveling of vehicles or walking as it occupies an installation space. Therefore, the charging apparatus is occasionally rendered movable, and housed under the ground, for example.

Meanwhile, a parking lot in which vehicles are parked occasionally includes parking spaces provided in a plurality of levels, such as when the parking lot is a multilevel parking lot or when parking spaces are provided both on and under the ground. Therefore, in order to enable electric vehicles to be charged in each level, it is conceivable to install the movable charging apparatus in each level, for example. However, it may be expensive to install the movable charging apparatus in each level.

Thus, in the present embodiment, as described with reference to FIGS. **1**, **2**, and **3**, the charging station **300** is a movable charging apparatus that is installed in the partition portion **104** which separates the first parking space **100** and the second parking space **102** and that can charge an energy storage device (the battery **214***a* or the battery **214***b*) mounted on an electric vehicle (the electric vehicle **200***a* or the electric vehicle **200***b*) parked in one of the first parking space **100** and the second parking space **102**. The charging station **300** includes the movable portion **300***a* that includes the connector **302** which is a connection device that is connectable to the battery **214***a* or the battery **214***b*, and the elevating device **306** that moves the movable portion **300***a* to one of the first position at which the movable portion **300***a* projects into the space on the first parking space **100** side with respect to the partition portion **104**, the second position at which the movable portion **300***a* projects into the space on the second parking space **102** side with respect to the partition portion **104**, and the third position at which the movable portion **300***a* is housed in the partition portion **104**.

With such a configuration, the battery mounted on the vehicle parked in either the first parking space **100** or the second parking space **102** can be charged using a single charging station **300**. Consequently, it is not necessary to install the charging station **300** in each of the first parking space **100** and the second parking space **102**. Therefore, it is possible to suppress an increase in the installation cost for installing the charging apparatus so as to be usable in a plurality of parking spaces.

An example of control processes executed by the control device **308** will be described below with reference to FIG. **4**. FIG. **4** is a flowchart illustrating an example of processes executed by the control device **308**. The sequence of processes indicated in the flowchart is executed by the control device **308** repeatedly in predetermined control cycles.

In step (hereinafter abbreviated as "S") **100**, the control device **308** determines whether the movable portion **300***a* is located at the third position. The control device **308** determines that the movable portion **300***a* is located at the third position when the third flag discussed above is in the on state, for example. The process proceeds to step S**102** when it is determined that the movable portion **300***a* is located at the third position (YES in step S**100**).

In step S**102**, the control device **308** determines whether there is an elevation request. Specifically, the control device **308** may determine that there is an elevation request when information that indicates a request to execute the elevation control is received from the electric vehicle **200***a*, the management server, a portable terminal carried by the user of the electric vehicle **200***a*, or an external device such as another charging station **300**, for example. The process proceeds to S**104** when it is determined that there is an elevation request (YES in S**102**).

In S**104**, the control device **308** executes the elevation control until the movable portion **300***a* reaches the first position. The process proceeds to S**106** when it is determined that there is no elevation request (NO in S**102**).

In step S**106**, the control device **308** determines whether there is a lowering request. Specifically, the control device **308** may determine that there is a lowering request when information that indicates a request to execute the lowering control is received from the electric vehicle **200***a*, **200***b*, the management server, a portable terminal carried by the user of the electric vehicle **200***a*, a portable terminal carried by the user of the electric vehicle **200***b*, or an external device such as another charging station **300**, for example. The process proceeds to S**108** when it is determined that there is a lowering request (YES in S**106**).

In S**108**, the control device **308** executes the lowering control until the movable portion **300***a* reaches the second position. This process is ended when it is determined that there is no lowering request (NO in S**106**). The process proceeds to S**110** when it is determined that the movable portion **300***a* is not located at the third position (NO in S**100**).

In S**110**, the control device **308** determines whether there is a storage request. Specifically, the control device **308** may determine that there is a storage request when information that indicates a request to execute the storage control is received from the electric vehicle **200***a*, **200***b*, the management server, a portable terminal carried by the user of the electric vehicle **200***a*, a portable terminal carried by the user of the electric vehicle **200***b*, or an external device such as another charging station **300**, for example. The process proceeds to S**112** when it is determined that there is a storage request (YES in S**110**).

In S**112**, the control device **308** determines whether the movable portion **300***a* is located at the first position. The control device **308** determines that the movable portion **300***a* is located at the first position when the first flag discussed above is in the on state, for example. The control device **308** determines that the movable portion **300***a* is not located at the first position when the second flag is in the on state, for example. The process proceeds to step S**114** when it is determined that the movable portion **300***a* is located at the first position (YES in step S**112**).

In S**114**, the control device **308** executes the lowering control until the movable portion **300***a* reaches the third position. The process proceeds to S**116** when it is determined that the movable portion **300***a* is not located at the first position (NO in S**112**).

In S**116**, the control device **308** executes the elevation control until the movable portion **300***a* reaches the third position. This process is ended when it is determined that there is no storage request (NO in S**110**).

An example of operation of the control device **308** according to the present embodiment based on the structure and the flowchart described above will be described.

A case where the movable portion **300***a* is located at the third position as illustrated in FIG. **1**, for example, is assumed. When the movable portion **300***a* is located at the third position (YES in S**100**), it is determined whether there is an elevation request (S**102**). When there is no elevation request (NO in S**102**), it is determined whether there is a lowering request (S**106**).

For example, when a request to elevate the charging station **300** is made by the user of the electric vehicle **200***a* parked in the first parking space **100** using the electric vehicle **200***a* or using the portable terminal carried by the user of the electric vehicle **200***a* (YES in S**102**), the elevation control is executed until the movable portion **300***a* reaches the first position (S**104**).

With the movable portion **300***a* moved to the first position, the user of the electric vehicle **200***a* can take the connector **302** out of the movable portion **300***a* and connect the connector **302** that has been taken out to the inlet **220***a* of the electric vehicle **200***a*. This allows the battery **214***a* mounted on the electric vehicle **200***a* to be charged.

Alternatively, when a request to lower the charging station **300** is made by the user of the electric vehicle **200***b* parked in the second parking space **102** using the electric vehicle **200***b* or using the portable terminal carried by the user of the electric vehicle **200***b* (YES in S**106**), for example, the lowering control is executed until the movable portion **300***a* reaches the second position (S**108**).

With the movable portion **300***a* moved to the second position, the user of the electric vehicle **200***b* can take the connector **302** out of the movable portion **300***a* and connect the connector **302** that has been taken out to the inlet **220***b* of the electric vehicle **200***b*. This allows the battery **214***b* mounted on the electric vehicle **200***b* to be charged.

Next, a case where the movable portion **300***a* is located at the first position as illustrated in FIG. **2**, for example, is assumed. It is determined whether there is a storage request (S**110**), since the movable portion **300***a* is not located at the third position (NO in S**100**).

When, because of a completion of charging the battery **214***a*, a request to house the charging station **300** is made by the user of the electric vehicle **200***a* using the electric vehicle **200***a* or the portable terminal carried by the user of the electric vehicle **200***a* (YES in S**110**), for example, the lowering control is executed until the movable portion **300***a* reaches the third position (S**114**), since the movable portion **300***a* is located at the first position (YES in S**112**). When the movable portion **300***a* is moved to the third position, the movable portion **300***a* is housed in the partition portion **104**.

Next, a case where the movable portion **300***a* is located at the second position as illustrated in FIG. **3**, for example, is assumed. It is determined whether there is a storage request (S**110**), since the movable portion **300***a* is not located at the third position (NO in S**100**).

When, because of a completion of charging the battery **214***b*, a request to house the charging station **300** is made by the user of the electric vehicle **200***b* using the electric vehicle **200***b* or the portable terminal carried by the user of the electric vehicle **200***b* (YES in S**110**), for example, the elevation control is executed until the movable portion **300***a* reaches the third position (S**116**), since the movable portion **300***a* is not located at the first position (NO in S**112**). When the movable portion **300***a* is moved to the third position, the movable portion **300***a* is housed in the partition portion **104**.

With the charging station **300** as the charging apparatus according to the present embodiment, as described above, it is possible to charge the battery mounted on the vehicle parked in either the first parking space **100** or the second parking space **102**, which are located separately in the upper and lower levels, using the single charging station **300**. Consequently, it is not necessary to install the charging station **300** in each of the level of the first parking space **100** and the level of the second parking space **102**. Thus, it is possible to provide a charging apparatus that suppresses an increase in the installation cost for installing the charging apparatus so as to be usable in a plurality of parking spaces.

Further, the movable portion **300***a* can be moved to any of the first position, the second position, and the third position in accordance with a request from an external device such as the electric vehicle **200***a*, **200***b*, the portable terminal of the user of the electric vehicle **200***a*, the portable terminal of the user of the electric vehicle **200***b*, or the management server that manages the charging station **300**.

Modifications will be described below. While the power source is described as an AC power source in the embodiment discussed above, the power source may be a DC power source. In this case, the chargers **212***a*, **212***b* may be omitted from the electric vehicles **200***a*, **200***b*, for example.

While the connector **302** is described as being housed in the storage space in the movable portion **300***a* in the embodiment discussed above by way of example, a socket may be provided as exposed on a side surface of the middle portion, in the up-down direction, of the movable portion **300***a*, for example. With such a configuration, the user can charge the battery **214***a* mounted on the electric vehicle **200***a* by connecting the socket of the charging station **300** and the inlet **220***a* of the electric vehicle **200***a*, or can charge the battery **214***b* mounted on the electric vehicle **200***b* by connecting the socket of the charging station **300** and the inlet **220***b* of the electric vehicle **200***b*, using a charging cable prepared separately.

While the housing of the charging station **300** is described as having a cylindrical shape in the embodiment discussed above by way of example, the housing may have any shape that specifically enables elevating operation, and is not specifically limited to having a cylindrical shape. For example, the housing of the charging station **300** may have a rectangular parallelepiped shape.

While the first parking space **100** and the second parking space **102** are located separately in the upper and lower levels and the movable portion **300***a* is rendered movable in the up-down direction to be able to charge the battery of the electric vehicle parked in either of the parking spaces using the single charging station **300** in the embodiment discussed above by way of example, the first parking space and the second parking space may be located adjacent to each other in the horizontal direction, for example.

The installation cost for a charging apparatus may also be high when a movable charging apparatus is installed for each of a large number of parking spaces set in a single-level parking lot. Therefore, the charging station **300** may be provided in a partition portion set between a partition wall of a first parking space and a partition wall of a second parking space.

FIG. **5** illustrates an example of the configuration of the charging station **300** in a state in which the movable portion **300***a* is housed in a partition portion **404** between a first parking space **400** and a second parking space **402** and the electric vehicles **200***a*, **200***b* according to a modification.

In the modification, as illustrated in FIG. **5**, the first parking space **400** and the second parking space **402** are located adjacent to each other in the horizontal direction via the partition portion **404**. More specifically, the first parking space **400** and the second parking space **402** are set such that the electric vehicle **200***a* parked in the first parking space **400** and the electric vehicle **200***b* parked in the second parking space **402** are arranged side by side with each other.

The charging station **300** has a housing in a cylindrical shape, and is installed in an opening portion **406** formed in a portion of the partition portion **404** between a partition wall **408** of the first parking space **400** and a partition wall **410** of the second parking space **402**.

The opening portion **406** is formed to penetrate the partition portion **404** in the horizontal direction. The opening portion **406** has a circular cross-sectional shape, and is formed to have a predetermined gap between an inner peripheral surface of the opening portion **406** and an outer peripheral surface of the housing of the charging station **300**. Further, the opening portion **406** is formed so as to have a length that is about the same as the length of the charging station **300** in the horizontal direction (the up-down direction on the sheet surface of FIG. **5**) in the state illustrated in FIG. **5**.

The charging station **300** includes a movable portion **300***a* and a fixed portion **300***b*. A storage space in which a connector **302** (see FIGS. **6** and **7**) can be housed is formed in the movable portion **300***a*. One end of a cable **304** (see FIGS. **6** and **7**) is connected to the connector **302**, while the other end of the cable **304** is connected to a power source (not illustrated). The configuration of the connector **302**, the cable **304**, and the power source is similar to the configuration of the connector **302**, the cable **304**, and the power source described in relation to the embodiment discussed above. Therefore, detailed description thereof will not be repeated.

The movable portion **300***a* of the charging station **300** is configured to be movable to a first position at which the movable portion **300***a* projects into a space on the first parking space **400** side.

FIG. **6** illustrates an example of the configuration of the charging station **300** with the movable portion **300***a* located at the first position and the electric vehicles **200***a*, **200***b* according to the modification. When the movable portion **300***a* of the charging station **300** is moved to the first position on the first parking space **400** side, as illustrated in FIG. **6**, the connector **302** housed in the movable portion **300***a* can be taken out of the movable portion **300***a* to be connected to an inlet (not illustrated) of the electric vehicle **200***a* parked in the first parking space **400**.

Further, the movable portion **300***a* is configured to be movable to a second position at which the movable portion **300***a* projects into a space on the second parking space **402** side.

FIG. **7** illustrates an example of the configuration of the charging station **300** with the movable portion **300***a* located at the second position and the electric vehicles **200***a*, **200***b* according to the modification. When the movable portion **300***a* of the charging station **300** is moved to the second position on the second parking space **402** side, as illustrated in FIG. **7**, the connector **302** housed in the movable portion **300***a* can be taken out of the movable portion **300***a* to be connected to an inlet (not illustrated) of the electric vehicle **200***b* parked in the second parking space **402**. Further, the movable portion **300***a* is configured to be held at a third position at which the movable portion **300***a* is housed in the partition portion **404**.

The fixed portion **300***b* is fixed to the inner wall portion of the opening portion **406**. The fixed portion **300***b* includes a moving device (not illustrated) that moves the movable portion **300***a* in the horizontal direction (the up-down direction on the sheet surface of FIG. **5**) and a control device (not illustrated) that controls operation of the moving device.

The moving device according to the modification is different from the elevating device **306** according to the embodiment discussed above in the direction of movement of the movable portion **300***a*, and is otherwise configured in the same manner as the elevating device **306** according to the embodiment discussed above. Therefore, detailed description thereof will not be repeated. The control device according to the modification is different form the control device **308** according to the embodiment discussed above in controlling the moving device discussed above in place of the elevating device **306**, and is otherwise configured in the same manner Therefore, detailed description thereof will not be repeated.

The control device controls the moving device such that the movable portion **300***a* is moved from the third position to the first position when a condition for executing first movement control is met, for example. The condition for executing the first movement control includes a condition that the movable portion **300***a* is located at the third position and a condition that there is a request (hereinafter referred to as a "first movement request") to execute the first movement control. The first movement request may be received from the electric vehicle **200***a* parked in the first parking space **400**, or may be received from the management server, the portable terminal, or another charging station **300**, for example. The condition for executing the first movement control may include a condition that the movable portion **300***a* is located at the second position or the third position, in place of the condition that the movable portion **300***a* is located at the third position. The control device detects the amount of movement of the movable portion **300***a* using a sensor, and moves the movable portion **300***a* until the amount of movement of the movable portion **300***a* reaches a value that is equivalent to the first position, for example.

Alternatively, the control device controls the moving device such that the movable portion **300***a* is moved from the third position to the second position when a condition for executing second movement control is met, for example. The condition for executing the second movement control includes a condition that the movable portion **300***a* is located at the third position and a condition that there is a request (hereinafter referred to as a "second movement request") to execute the second movement control. The second movement request may be received from the electric vehicle **200***b* parked in the second parking space **402**, or may be received from the management server, the portable terminal, or another charging station **300**, for example. The condition for executing the second movement control may include a condition that the movable portion **300***a* is located at the first position or the third position, in place of the condition that the movable portion **300***a* is located at the third position. In addition, the control device moves the movable portion **300***a* until the amount of movement of the movable portion **300***a* reaches a value that is equivalent to the second position, for example.

Further, the control device controls the moving device such that the movable portion **300***a* is moved from the first position or the second position to the third position when the condition for executing the storage control is met. Specifically, the control device executes movement control such that the movable portion **300***a* is moved from the first position to the third position to be housed in the partition portion **404** when the movable portion **300***a* is located at the first position and the condition for executing the storage control is met, for example. Further, the control device executes movement control such that the movable portion **300***a* is moved from the second position to the third position to be housed in the partition portion **404** when the movable portion **300***a* is located at the second position and the condition for executing the storage control is met, for example. In the movement control, the control device moves the movable portion **300***a* until the amount of movement of the movable portion **300***a* reaches a value that is equivalent to the third position.

The condition for executing the storage control includes a condition that the movable portion **300***a* is located at the first position or the second position and a condition that there is a storage request to house the charging station **300**. The storage request may be received from the electric vehicle **200***a* or the electric vehicle **200***b* when the movable portion **300***a* is located at the first position, for example. Alternatively, the storage request may be received from the electric vehicle **200***a* or the electric vehicle **200***b* when the movable portion **300***a* is located at the second position, for example. Alternatively, the storage request may be received from the management server, the portable terminal, or another charging station **300**, for example.

With such a configuration, the control device executes the processes indicated in FIG. **8**. FIG. **8** is a flowchart illustrating an example of processes executed by the control device according to the modification. The "first movement request", the "second movement request", the "first movement control", and the "second movement control" in the flowchart in FIG. **8** correspond to the "elevation request", the "lowering request", the "elevation control", and the "lowering control", respectively, in the flowchart in FIG. **4**. Therefore, the processes in **S200**, **S202**, **S204**, **S206**, **S208**, **S210**, **S212**, **S214**, and **S216** in FIG. **8** are substantially the same as the processes in **S100**, **S102**, **S104**, **S106**, **S108**, **S110**, **S112**, **S114**, and **S116**, respectively, in FIG. **4** except that the movable portion **300***a* is moved in the horizontal direction, rather than in the vertical direction. Therefore, detailed description of the content of the processes indicated in the flowchart in FIG. **8** and the operation of the control device will not be repeated.

Also with such a configuration, the battery mounted on the vehicle parked in either the first parking space or the second parking space, which are adjacent in the horizontal direction, can be charged using a single charging station **300**. Consequently, it is not necessary to install the charging apparatus in each of the first parking space and the second parking space. Therefore, it is possible to suppress an increase in the installation cost for installing the charging apparatus so as to be usable in a plurality of parking spaces.

The modifications described above may be wholly or partially combined with each other as appropriate. The embodiment disclosed herein should be construed as illustrative in all respects and not restrictive. The scope of the present disclosure is defined by the claims, rather than the above description, and is intended to include all modifications that fall within the meaning and scope of equivalence to the claims.

What is claimed is:

1. A charging apparatus for charging an energy storage device mounted on a vehicle comprising:

a movable portion that includes a connection device that is connectable to the energy storage device of the vehicle parked in one of a first parking space or a second parking space; and

a moving device configured to move the movable portion to one of a first position at which the movable portion projects into a space that is close to the first parking space with respect to a partition portion that separates the first parking space and the second parking space, a second position at which the movable portion projects into a space that is close to the second parking space with respect to the partition portion, and a third position at which the movable portion is housed in an opening portion in the partition portion, wherein:

the charging apparatus is installed in the opening portion so that outer peripheral surface of the charging apparatus is away from an inner peripheral surface of the opening portion by a predetermined gap,

wherein the first parking space is located in a level directly above a level of the second parking space, and the partition portion is provided between a floor of the first parking space and a ceiling of the second parking space, and

the first parking space is located in a level directly above a level of the second parking space, and the partition portion is between a floor of the first parking space and a ceiling of the second parking space.

2. The charging apparatus according to claim **1**, wherein the first parking space is adjacent to the second parking space in a horizontal direction, and the partition portion is provided between a partition wall of the first parking space and a partition wall of the second parking space.

3. The charging apparatus according to claim **1**, further comprising a control device configured to control the moving device using information acquired from an external device that is external to the charging apparatus, wherein the control device is configured to control the moving device so as to move the movable portion to one of the first position, the second position, and the third position corresponding to a movement request to move the movable portion when the movement request is acquired from the external device.

4. The charging apparatus according to claim **1**, wherein the moving device includes a sensor.

5. The charging apparatus according to claim **4**, wherein the moving device is configured to:

detect an amount of elevation of the movable portion by the sensor, and

elevate the movable portion until the amount of elevation of the movable portion reaches a predetermined value for the movable portion to reach the first position.

6. A charging apparatus for charging an energy storage device mounted on a vehicle comprising:

a movable portion that includes a connection device that is connectable to the energy storage device of the vehicle parked in one of a first parking space or a second parking space; and

a moving device configured to move the movable portion to one of a first position at which the movable portion projects into a space that is close to the first parking space with respect to a partition portion that separates the first parking space and the second parking space, a second position at which the movable portion projects into a space that is close to the second parking space with respect to the partition portion, and a third position at which the movable portion is housed in an opening portion in the partition portion, wherein:

the charging apparatus is installed in the opening portion so that outer peripheral surface of the charging apparatus is away from an inner peripheral surface of the opening portion by a predetermined gap,

wherein the first parking space is located in a level directly above a level of the second parking space, and the partition portion is provided between a floor of the first parking space and a ceiling of the second parking space,

wherein the first parking space is located in a level directly above a level of the second parking space, and the partition portion is between a floor of the first parking space and a ceiling of the second parking space.

7. The charging apparatus of claim **6**, wherein the first position is above the floor of the first parking space.

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